MIL-D-2273E
13 July 1979
SUPERSEDING
MIL-D-2273D
22 December 1966

MILITARY SPECIFICATION

DYE MIX: (RECOLORING AND FIELD-DYEING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of dye mix for recoloring previously colored items and one type of a dye mix for field dyeing of white items (see 6.1).

1.2 Classification. The dye mix covered by this specification shall be of one grade and of the following types as specified (see 6.2).

Type I	-	Dye Mix:	Re-Coloring, Cotton - Dark Blue 351
Type II	-	Dye Mix:	Re-Coloring, Wool, Cotton - Blue 352
Type III	-	Dye Mix:	Re-Coloring, Wool, Cotton - Brown 353
Type IV	-	Dye Mix:	Field Dyeing, Cotton - OG 109
Type V	-	Dye Mix:	Re-Coloring - Blue Green No. 320

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6820

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-P-704 - Pails, Metal: (Shipping, Steel 1 Through 12 Gallon).
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.
PPP-T-76 - Tape, Packaging, Paper (For Carton Sealing).

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard sample. Equal weights of the dye mix and of the standard dye mix sample applied to the same weight of the same fabric under identical conditions shall produce the same shade (i.e., same lightness and chromaticity). The standard dye mix sample and the submitted dye mix shall have essentially the same exhaustion rates, and the submitted dye mix shall be equal to or better than the standard dye mix sample in regards to colorfastness properties (see 3.4 and 6.3). The following percentages of the standard dye mix applied under the conditions further delineated in this specification match the end use shades for each type. The percentages shall be used in developing the samples for colorfastness tests from the standard dye mix sample and from the submitted dye mix.

Type I - 4.0
Type II - 4.0
Type III - 3.0
Type IV - 1.95
Type V - 1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

3.2 First article. When specified (see 6.2) before production is commenced, a sample of dye mix of the specified type shall be submitted to the contracting officer or his authorized representative for examination and approval in accordance with 4.3. Approval of first article sample authorizes commencement of production, but does not relieve the contractor of responsibility for complying with all the requirements of this specification. The first article sample shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

3.3 Material.

3.3.1 Types I, II, III, and V.- The dye mix shall be furnished as a homogeneous mixture, i.e., all components blended into one unit, each sample of which in a random sampling of a container or a lot shall provide the same lightness and chromaticity values. The submitted dye mix at the same concentration level as the standard dye mix shall provide the same shade, tinctorial strength and other characteristics as the reference dye mix standard for the applicable type. Each unit pack shall have the same total dyeing capacity as the dye contained in a pack of the strength of the referenced standard dye mix applied to the same weight of cloth or clothing.

3.3.2 Type IV. The type IV dye mix shall be furnished in an intermediate pack containing three separate component plastic bags, one holding 4 lbs (1.8 kg) of common salt another 3.2 oz (91 g) of a cationic water soluble aminoplast, free of formaldehyde, for after-fixing the dye and a bag containing 6-1/4 ozs (177 g) of the dye mix of the same tinctorial strength as the reference dye mix sample or, in the event the submitted dye mix is stronger tinctorially than the standard dye mix sample, the dye pack may weigh proportionally less, but the total dyeing capacity when applied to 20 lbs (9 kg) of cloth or clothing shall be the same as would be attainable with 6-1/4 ozs (177 g) of the referenced dye mix standard applied to 20 lbs (9 kg) of cloth or clothing.

3.4 Colorfastness. Fabric dyed with the dye mixes in accordance with the methods described in 4.5.1 to the depth of shade achieved by the respective type standard dye mix samples applied at the percentage levels (see 3.1) shall show colorfastness to light, laundering and perspiration equal to or better than is shown by the fabric dyed with the applicable standard dye mix sample to the same lightness level when tested as specified in 4.5.2.

3.5 Salt (type IV). The salt, (sodium chloride), to be provided with type IV dye mix and any used for standardizing the dye mix to the tinctorial strength of the reference dye mix standard shall be that commercially designated as common salt, purified, calcium-free. For standardizing, sodium sulfate calcined, calcium-free, may be substituted for the sodium chloride, but such substitution shall be annotated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

3.6 Instructions for use (type IV only). The following printed instructions shall be included with each intermediate pack of type IV dye mix.

INSTRUCTIONS FOR USE OF TYPE IV DYE MIX, OG 109

1. The dyeing shall be accomplished in mobile laundries, fixed laundry installations, or may be accomplished in any suitable container available in the field.

a. The following general procedures are applicable for use in fixed laundry equipment. The dyeing shall be carried out in 20-pound (9.0 kg) units of dry clothing, or in multiples thereof, depending upon the size of the wash wheel. Table I contains data for different wash wheels in respect to size, water level, and quantity of dye and salt to be used. The ratio shall be 1 package of dye for each 20 pounds (9.0 kg) of clothing to one 4 pound (1.8 kg) package of salt for each 20 pounds (9.0 kg) of clothing (30 undershirts, 30 shorts, 10 towels, 20 handkerchiefs and 10 face cloths weigh 20 pounds (9.0 kg).

(1) Dissolving the dye.

(a) Place the dye mix in a 5-gallon (19 L) container using 1 package of dye for each 20 pounds (9.0 kg) of clothing items to be dyed.

(b) Add sufficient cold water to the dyes in the container while stirring to make a paste and all dye powder is thoroughly wetted.

(c) Slowly pour boiling water into the dye paste while stirring until 5 gallons (19 L) of dissolved dye solution is made in the container.

(2) Dyeing procedure.

(a) Load the machine with clothing to be dyed.

(b) Add 8 inches of cold water and run 5 minutes.

(c) Add the dissolved dyes into the running washer through the slots taking care not to spill dye solution.

(d) Add one half the amount of common salt required for the load while machine is running.

(e) Heat the dyebath in the washer to 190°F (88°C) in 30 minutes. Add remaining half of common salt required for the load.

(f) Run the washer at 190°F (88°C) for 10 minutes more.

(g) Shut steam, open slot and run for 10 minutes to allow bath to cool.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

(h) Drain.

(i) Add cold water to 14-inch (356 mm) level and run 5 minutes.

(j) Drain.

(k) Add cold water to 14-inch (356 mm) level and run 5 minutes.

(l) Drain.

(m) Add cold water to 10-inch (254 mm) level.

(n) Add after-fixing compound 1/ using 3.2 oz (91 g) package for each 20 lbs (9.0 kg) of clothing items.

(o) Raise temperature to 130°F (54°C) in 10 minutes.

(p) Run at 130°F (54°C) for 15 minutes.

(q) Rinse.

(r) Drain. Add water to 10-inch (254 mm) level and run for 5 minutes.

(s) Drain.

(t) Hydroextract dry.

1/ After-fixing compound should be dissolved by sprinkling into 130°F (54°C) water in a pail or similar container while stirring and added to wash wheel as a solution, not as dry powder.

NOTE: Steps (a) through (j) above require one hour. Equipment utilizing cycle charts will require two charts, one for steps (a) to (j), the second for the remaining steps. Chart will be changed after step (j).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

TABLE I. Wash wheel size and dyeing materials for fixed laundry equipment

Installation (wash wheel size) (inches)	Normal operating level of water (inches)	Weight of dry material (pounds)	Number unit package of dye	Common salt (pounds)	Water volume to dissolve mixed dye (gallons)
30 x 30 (762 mm 762 mm)	8 (203 mm)	80 (36 kg)	4	16 (7 kg)	2 (8 L)
36 x 54 (914 mm x 1372 mm)	8 (203 mm)	140 (64 kg)	7	28 (12 kg)	5 (19 L)
42 x 72 (1067 mm x 1829 mm)	8 (203 mm)	200 (91 kg)	10	40 (18 kg)	5 (19 L)
42 x 84 (1067 mm x 2134 mm)	8 (203 mm)	300 (136 kg)	15	60 (27 kg)	5 (19 L)
42 x 96 (1067 mm x 2438 mm)	8 (203 mm)	340 (154 kg)	17	68 (31 kg)	5 (19 L)

b. The following general procedures are applicable for use in a single trailer mounted laundry (the load for the single trailer laundry unit is 60 lbs (27.2 kg)); for this purpose, charts may be cut. Two charts will be required as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

CHART No. 1 DYEING CYCLE - Set aquaset at 4.5 to deliver water at 180°F (82°C) or hotter

Operation	Water level (inches)	Water temperature	Time	Supplies
Pre-wet	8 (203 mm)	180°F plus (82°C)	4 minutes	None
Drain	--	--	1/2 minute	None
Dyeing	11 (high level) (279 mm)	180°F plus (82°C)	--	3 dye packets
Dyeing	--	Hot	20 minutes	1/2 salt
Dyeing	--	Hot	10 minutes	1/2 salt
Dyeing	--	Hot	10 minutes	--
Drain	--	--	1/2 minute	--
Pre-extract	--	--	1/2 minute	--
Brake	--	--	1/2 minute	--
Extract	--	--	2 minutes	--
Rinse	8 (low level) (203 mm)	Cold	5 minutes	--
Drain	--	--	1/2 minute	--
Rinse	8 (low level) (203 mm)	Cold	5 minutes	--
Drain	--	--	1/2 minute 59 minutes	--

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

CHART No. 2

Operation	Water level (inches)	Water temperature	Time	Supplies
Rinse	8 (low level) (203 mm)	Cold	4 minutes	None
Drain	--	--	1/2 minute	--
Rinse	8 (203 mm)	Cold	4 minutes	None
Drain	--	--	1/2 minute	--
Aftertreat	10 (254 mm)	130°F (54°C)	20 minutes	3 packets aftertreat
Drain	--	--	1/2 minute	--
Pre-extract	--	--	1/2 minute	None
Brake	--	--	1/2 minute	--
Extract	--	--	6 minutes	--
			31 minutes	

Dry in 30 lb (13.6 kg) units (13 to 18 minutes) with stack temperature at 150° to 160°F (66° to 71°C).

(1) Dissolving the dye.

(a) Place the dye mix in a 5 gallon (19 L) container using 1 package of dye for each 20 pounds (9 kg) of clothing items to be dyed.

(b) Add sufficient cold water to the dyes in the container while stirring to make a paste and all dye powder is thoroughly wetted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

(c) Slowly pour boiling water into the dye paste while stirring until 5 gallons (19 L) of dissolved dye solution is made in the container.

(2) After-fixing compound should be dissolved by sprinkling into 130°F (54°C) water in a pail or similar container while stirring and added to wash wheel as a solution, not as dry powder.

c. Improvised equipment. If the dyeing is to be accomplished with improvised equipment, the amount of water shall be 2-1/2 gallons (9.5 L) for each pound of clothing to be dyed. The following general procedures are applicable for use in the field application of type IV dye mix, OG 109, by small groups or by individuals. Dyeing of white cotton undershirts, shorts, towels and handkerchiefs can be accomplished in any available metal container, preferably a 55-gallon (208 L) steel drum with open end, which is capable of withstanding heating over fire or by other available methods.

PROCEDURE FOR DYEING 20 POUNDS (9 kg) 1/

1. Fill metal container (steel drum) half full with water from nearby source (stream, lake, river, etc.) or other available sources and start heating.

2. Place contents of one dye packet (dye only) in a metal container 2-gallon (7.5 L) capacity such as empty ration cans, pails, helmet or other suitable container.

3. Slowly add cold water to the dye powder while stirring with a stick or other device to make a paste.

4. Pour hot water over the paste while stirring to dissolve the dye.

5. Drop clothing items into the drum of water while stirring and turn the items to wet them completely.

6. Slowly add dissolved dyestuffs to the drum while stirring and turning the clothing items until all the dye solution has been added.

7. Add half the package of salt to the dyebath while stirring the items.

8. Continue heating the drum, while stirring the items, until the temperature of the dyebath is close to a boil.

9. Add the remaining half quantity of salt and continue stirring and dyeing at a near boil of 30 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

10. Remove source of heat and continue dyeing, while stirring, in a cooling bath for 20 minutes.

11. Remove dyed items or dump dyebath and rinse dyed items in fresh clean cold water several times until rinse waters are reasonably clear or free of color from the dyes.

12. Half fill drum with fresh warm water heated to lukewarm 130°F (54°C).

13. Sprinkle into the warm water the after-fixing compound while stirring until all is well dissolved.

14. Add the well rinsed dyed items to the after-mix solution in the drum and stir items for 15 minutes.

15. Remove dyed items.

16. Rinse well.

17. Squeeze out excess water.

18. Dry.

1/ Items totaling 20 pounds (9 kg) consist of 30 undershirts, 20 shorts, 10 towels, 20 handkerchiefs and 10 face cloths which are the standard issue to 5 troops.

3.7 Workmanship. The finished product shall be uniform and free from foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection.

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. Inspection of the first article sample shall be conducted to determine compliance with all the requirements of this specification.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated. For purposes of sampling, a lot shall consist of all material from the same manufactured batch.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. Additional component testing shall be performed in accordance with table II. The lot shall be unacceptable if any sample does not meet the requirements of any characteristic.

TABLE II. Instructions for testing components and materials

After-fixing agent (type IV only)	3.3.2 1/
Sodium chloride (type IV only)	3.5 2/

1/ Trade name and certificate of compliance shall be furnished for this item.

2/ Certificate of compliance shall be furnished for this item.

4.4.2 End item inspection. The end item shall be examined for the defects in the applicable subparagraphs. The inspection level (sample size) shall be S-3 and the acceptable quality level (AQL) shall be 2.5 defects per 100 units. The lot size shall be expressed in unit containers of the specified capacity for the examination in 4.4.2.1.1 and 4.4.2.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

4.4.2.1 End item examination.

4.4.2.1.1 Visual examination. The sample unit for this examination shall be the contents of one filled unit container.

Examine	Defect
Workmanship	Not uniform, dirty, presence of foreign matter.
Components (type IV only)	Salt missing, dye missing, after-fixing compound missing, instruction sheet missing. Components not positioned as specified. Instruction sheet incomplete, incorrect. Component polyethylene bags not as specified, not sealed as specified; excess air not expelled (evidence of pillow effect caused by trapped excess air). Polyethylene bags not marked as specified (see 5.3.2).

4.4.2.1.2 Examination for average net content.

4.4.2.1.2.1 Types I, II, III, and V. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container is less than specified for all sample units examined.

4.4.2.1.2.2 Type IV. The sample unit for this examination shall be the salt, dye and after-fixing compound components of the unit container. The individual unit pack of salt shall contain a minimum of 4 pounds (1.8 kg). The individual unit pack of the after-fixing compound (dry powder) shall weigh approximately 3.2 ozs (91 g). The unit pack of dye mix shall weigh approximately 6-1/4 ozs (177 g). If the dye mix offered is tinctorially stronger than the reference standard dye mix sample, the unit pack of dye mix may weigh proportionally less, but the total dyeing capacity shall be equal to that of 6-1/4 ozs (177 g) of the standard dye mix sample.

4.4.2.2 End item testing. The end item shall be tested for the characteristics specified in 3.4 by the method specified in FED-STD-191, where applicable, and in table III. The lot size shall be expressed in pounds for types I, II, III and V and in the number of intermediate packs for type IV. The sample unit for testing types I, II, III, and V shall be a one pound (453 g) composite obtained by combining equal portions selected at random throughout the lot and thoroughly mixed to yield the composite. The sample unit for testing type IV shall be one intermediate pack (including the salt, dye, and fixation agent). The sample size for tests shall be as indicated in the tables below. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

composite shall be placed in a clean, dry, sealed container. Care shall be exercised to prevent contamination or alteration of the composite during sampling, compositing, storage and testing. The lot shall be unacceptable if the sample unit or composite fails to meet any test requirement specified. The test cloths shall be prepared and evaluated as specified in 4.5.1 and 4.5.2. All tests shall be performed in accordance with standard conditions as defined in FED-STD-191. All test reports shall contain the individual values utilized in expressing the final result.

Lot size - types I, II, III, V (pounds)	Samples for composite
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Lot size - type IV (intermediate packs)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

TABLE III. End item tests

Characteristic	Specification reference		Requirements applicable to		Number	Results determi-	reported as	
	Reqmt	Test method	Individual unit type IV	Composite types I, II, III, V	per unit	nations per unit	Pass or fail	Numeric-ally to nearest
Tinctorial strength	3.3.1 and 3.3.2	4.5.1	X	X	1		See 4.5.1	
Colorfastness:								
To light	3.4	4.5.2 and 5660	X	X	1	X		
To laundering	3.4	4.5.2 and 5614	X	X	1	X		
To perspiration	3.4	4.5.2 and 5680	X	X	1	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

4.4.3 Packaging inspection. An examination shall be made to determine that preservation, packing and markings comply with section 5 requirements. The sample unit shall be one shipping container, fully packaged, selected just prior to the closing operation. The inspection level (sample size) shall be S-1 and the acceptable quality level (AQL) shall be 2.5 defects per 100 units. Closed shipping container shall be examined for closure defects.

Examine	Defect
Packaging	
Unit container (type IV)	Components not unit packed separately. Intermediate pack container not as specified. Closure not heat-sealed, leaks. Interior box not as specified. Metal fasteners used for securing fiberboard boxes. Closure not as specified.
Packing (type I, II, III, and V)	Container not as specified. Bulged or distorted container. Gaskets or seals omitted or defective. Closure not tight, leaks.
Packing (type IV)	Container not as specified. Closure, waterproofing and reinforcing, not accomplished by specified methods. Number of intermediate packs per shipping container is less than specified.
Markings	Incomplete, incorrect, or omitted. Not in accordance with requirements. Does not contain legend "SEE DYEING INSTRUCTIONS INSIDE" (type IV only).

4.5 Test methods.

4.5.1 Determination of tinctorial strength.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

4.5.1.1 Preparation of test specimens. The test specimens for types I, IV, and V dye mix shall be 10 gram samples of white 80 by 80, bleached cotton print cloth, style 400 of Testfabrics, Inc. or equal (see 6.5). The test specimens for types II and III dye mix shall be 10 gram samples of natural worsted flannel, 8 ounces per linear yard, style 523 of Testfabrics, Inc. or equal (see 6.5). A one-gram sample of bleached cotton print cloth shall be sewn to the worsted flannel.

4.5.1.2 Procedure. A 10 gram sample of the applicable fabric shall be thoroughly wet out and introduced into a dyebath containing the following percentages of the standard dye mix based on fabric weight.

Type I - 0.3 percent on weight of fabric.
Type II - 0.5 percent on weight of fabric.
Type III - 0.3 percent on weight of fabric.
Type IV - 0.6 percent on weight of fabric.
Type V - 1.0 percent on weight of fabric.

These percentages have been selected for ease in determining tinctorial equivalence by either visual or colorimetric techniques. Dyeings at varying percentages of the submitted dye mix on the weight of the fabric shall be made on 10 gram samples of the same fabric as used for dyeing the standard dye mix sample to establish the tinctorial equivalence values of the submitted dye mix to the standard reference dye sample. The ratio by weight of dye liquor to fabric shall be 30 to 1. The dyeing in each instance shall be started at 100°F (38°C) and the temperature raised to the boil in 30 minutes. When the boiling temperature has been reached, 20 percent on the weight of common salt shall be added to the dyebath and the dyeing continued at the boil for an additional 45 minutes with suitable agitation to assure uniform application of the dye. After the 45 minute boil, the dyebath shall be permitted to cool down and the fabric removed from the dyebath when the temperature has lowered to 160°F (71°C). The sample of dyed fabric shall be washed in successive baths, ratio 30 to 1, first in two (2) baths of 10 percent common salt solution and then in cool water until the rinse water is clear. The fabric shall be extracted by any suitable method, smoothed out, and dried at a temperature not exceeding 160°F (71°C). The test specimens dyed with the submitted dye mix shall be prepared as specified above. The cloth dyed with type IV dye mix shall not be aftertreated for the purposes of this test.

4.5.1.3 Evaluation. The dyed and conditioned test specimen prepared from the submitted dye mix and one from the standard samples shall be evaluated visually by the method specified in 4.5.1.3.1 or colorimetrically by using either a spectrophotometer or a Tristimulus colorimeter by the methods specified in 4.5.1.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

4.5.1.3.1 Visual procedure for matching tinctorial strength. The shade of the specimen dyed with the submitted dye mix shall match the shade of the sample dyed with the standard sample under artificial daylight having a color temperature of 7500° kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° kelvin.

4.5.1.3.1.1 Report. The report shall consist of a statement of the percent of the dye mix to be furnished calculated on the weight of fabric which is tinctorially equal to the defined percent dyeing of this reference standard dye mix (see 4.5.1.2). For types I, II, III, and V unless the percentages in the submitted dye mix and for the reference dye mix are the same (plus or minus 5 percent in lightness) the submitted dye mix shall be reported as failing. For the type IV, the report shall further define the quantity of dye which will be contained in the dye package for a total clothing capacity for 20 lbs (9 kg) of cloth or clothing equal to 6-1/4 ozs (177 g) of the standard dye mix sample.

4.5.1.3.2 Colorimetric procedure. The two specimens dyed with the submitted dye mix and the specimen dyed with the standard sample shall be evaluated in accordance with standard colorimetric practice. Measurements shall be taken on all specimens at:

Type I (a) 460 mu and (b) 610 mu
Type II (a) 450 mu and (b) 600 mu
Type III (a) 480 mu and (b) 610 mu
Type IV (a) 430 mu and (b) 605 mu
Type V (a) 445 mu and (b) 620 mu

From the reflectance R_a and R_b the K/S value will be calculated and then the functions $K/S - (K/S)_o$ shall be determined where K/S_o is the value for the white undyed fabric.

4.5.1.3.2.1 Calculation. The "tinctorial" equivalence (absorbency)" (Ae) value at each wave length shall be obtained by the following:

$$Ae = \frac{K/S - (K/S)_o}{C}$$

Where C is the percent of dye applied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

4.5.1.3.2.2 Report. The average Ae value of the two specimens prepared from the submitted sample shall be obtained and the percent tinctorial equivalence obtained the following equation:

$$\text{Percent tinctorial equivalence} = \frac{\text{Average Ae submitted sample}}{\text{Ae of standard sample}} \times 100$$

The tinctorial equivalence of the submitted sample at each wave length shall be reported to the nearest 0.1 percent.

4.5.2 Determination of colorfastness. The test specimens specified in 4.5.1.1 for the applicable type of dye mix used in determining colorfastness shall be dyed as specified in 4.5.1.2 to the depth of shade as indicated in 3.1. The test specimens shall be evaluated individually against dyes of the applicable standard sample when tested for fastness to laundering, Method 5614; light, Method 5660; and perspiration, Method 5680, of FED-STD-191. The results shall be reported by comparison of the test specimens with these from the standard dye mix. The report shall state pass or fail.

4.6 Test cloths. The test cloths specified in 4.5.1.1 for the applicable type shall be dyed using the procedure specified in 4.5.1.2 except that type IV shall be aftertreated with 1 percent of the aftertreating compound approved by the contracting officer at 140°F (60°C) for 15 minutes and subsequently rinsed.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type IV dye mix. Each component of the type IV dye shall be separately unit packed in the quantity specified in 3.3.2 within a close-fitting clear polyethylene film bag of 0.003 inch thickness (± 20 percent tolerance). The polyethylene bag shall be formed with heat-sealed seams that are straight and continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat-sealed. Prior to or during the final heat closure operation, excess air within the bag shall be expelled. The bag may be fabricated from polyethylene film tubing or sheeting. One complete dye mix (consisting of common salt, after-fixing material, and dye mix) shall be intermediately packed in a snug-fitting folding or set-up paperboard box conforming to variety 2, style III, type G, class 1 of PPP-B-566 or type I, variety 2, class A, style 4 of PPP-B-676, respectively; or a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, class weather-resistant, grade W5c or W5s of PPP-B-636, except that metal fasteners shall not be used for securing the manufacturers' joint. The dye mix components shall be neatly arranged within the interior box by placing the bag of salt on the bottom with the bagged after-fixing material and dye mix placed on top. The printed instructions specified in 3.6 shall then be placed within the box. Each interior box shall be securely closed with 2-inch (51 mm) minimum width tape conforming to PPP-T-76 or type II, grade B of PPP-T-45, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

5.1.2 Commercial. The type IV dye mix shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Types I, II, III, and V. Twenty-five pounds (11.3 kg) of dye mix of one type only, shall be packed in a pail conforming to type II, class 4 of PPP-P-704.

5.2.1.2 Type IV. Twelve type IV dye mixes, preserved as specified in 5.1, shall be packed in shipping containers conforming to overseas type of PPP-B-601. Closure and strapping shall be in accordance with the container specification.

5.2.2 Level B packing.

5.2.2.1 Types I, II, III, and V. Twenty-five pounds (11.3 kg) of dye mix of one type only, shall be packed as specified in 5.2.1.1.

5.2.2.2 Type IV. Twelve type IV dye mixes, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class weather-resistant, grade V3c, V3s, or V4s of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

5.2.3 Commercial packing. Dye mixes, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, unit packs, intermediate packs, and shipping containers shall be in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Special marking (type IV only). The top of each interior box shall be plainly marked in capital letters not less than 1/4 inch (6 mm) in height as follows:

"SEE DYEING INSTRUCTIONS SHEET INSIDE"

5.3.2 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

NOMENCLATURE
NET WEIGHT

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Types I, II and III dye mixes are intended for use in recoloring previously colored clothing items. Type IV mix is intended for use in field dyeing of white handkerchiefs, towels and cotton summer underwear. Type V dye mix is intended for use in recoloring hospital items. Type V is a substitute dye for C.I. Direct Green 14 which is no longer available.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) First article sample approval when required (see 3.2).
(d) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Standard sample. The standard samples for each type are available from Commander, Defense General Supply Center, ATTN: Dir/Technical Operations, Richmond, VA 23219.

6.4 Note to contracting officer. To insure that the amount of dye mix contained in the individual package units of type IV mix submitted by the contractor is sufficient to dye 20 lbs (9 kg) of fabric, the following calculation may be made:

$$\text{Net weight of submitted dye mix (ozs)} = \frac{6.25 \times \text{percent reported by contractor (see table III)}}{1.95}$$

6.5 Test fabric. The test fabric is available from Testfabrics, Inc., 200 Blackford Avenue, Middlesex, NJ 08846.

6.6 Dye mix formulations. Suggested but not mandatory formulations are listed below. The formulations have been compounded without the use of benzidene type dyes.

Type I Direct Blue 3RRL (Direct Blue 67)
Direct Blue BH-NB

Type II Acid Blue 5R (Acid Blue 113)
Alizarine Fast Blue AP (Acid Blue 25)
Intralite Blue 3RRL (Direct Blue 67)
Direct Blue BH-NB

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

Type III Direct Brown BR-NB
Direct Red 6BL (Direct Red 79)
Direct Black GX-NB
Acid Milling Red SWB (Acid Red 114)

Type IV Direct Fast Blue FF (Direct Blue 71)
Direct Orange TGLL (Direct Orange 61)
Direct Orange ERL (Direct Orange 37)

Type V Direct Green 2B-NB
Direct Blue 2B-NB
Direct Black GX-NB

6.7 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in U.S. customary units shall be controlling.

6.8 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - MC
Air Force - 68

Preparing activity:

Army - GL

Project No. 6820-0078

Review activities:

Army - MD
DLA - GS

User activities:

Army - AR

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-2273E
DYE MIX: (RECOLORING AND FIELD-DYEING)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760
OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRDNA-ES

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD-314



Commander
US Army Natick Research and Development Command
ATTN: DRDNA-ES
Natick, MA 01760

NATICK OP 1 AUG 78
FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:38Z
Check the source to verify that this is the current version before use.